

Under five minutes, **ten slides.**

What to say on every slide, **29 September.**

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Run of show · 29 Sept

Under five minutes, ten slides: 460 spoken words and 15 clicks, 3:38 at 150 words a minute. Detail that is not on a slide is in the notes here. The spine: **shown, felt, lived** (02, 03, 04), then **the map, why it compounds, the product** (05, 06, 07), then **who else, the money, the ask** (08, 09, 10). 08 is the fastest slide: one beat.

SLIDE	CLOCK	SEC	CLICKS	BEAT	ONE LINE TO LAND
01 Control is an illusion	0:00–0:13	13	1	hook	The map every cell lab is missing.
02 One lab, or six?	0:13–0:46	33	3	shown	The model isn't the hard part. The map is.
03 Do my cells look normal?	0:46–1:11	25	1	felt	Cells are the input nobody certifies.
04 Founder	1:11–1:36	25	2	lived	This is the thing I kept needing and nobody sold.
05 The map	1:36–1:54	18	2	the map	Where you sit predicts what you measure.
06 Every lab that joins sharpens the map	1:54–2:13	19	1	why it compounds	The labs that build it use it free.
07 The product	2:13–2:36	23	1	what	Send your controls. See where you stand.

SLIDE	CLOCK	SEC	CLICKS	BEAT	ONE LINE TO LAND
08 Identity, yes. Behaviour, no.	2:36– 2:46	10	1	who else	The other half of the certificate.
09 Eighteen months, four milestones	2:46– 2:58	12	1	the money	A first line that pays.
10 Every plate, placed in the field	2:58– 3:16	18	0	vision, ask	Have images of U2OS cells? Talk to me.

Captions stay hidden while you present (Tweaks › captions); the shared link and PDF show them. If you're late, skip the second sentence of 03 and the first sentence of 10.

01 Control is an illusion (0:00–0:13)

On screen while you're introduced. Two lines drop from the top like dive lines: green is the field's datum, pink is yours. Say the one sentence.

0:00 Hi, I'm Kat. Every result in biology is a difference from a control, and nobody checks the control. Datum Line is the map every cell lab is missing. *Click.*

02 One lab, or six? (0:13–0:46, three views)

Shown. The first view is a game: let the room look for four seconds.

0:13 Six untreated wells. Same line, same protocol, no drug. One lab, or six? *Click.*

0:20 Six. Nobody here can tell. A classifier can: 99.9 percent across eleven labs, against nine by chance, on plates it never saw. *Click.*

0:31 Why now: this data is public and free, and models are a commodity. The model isn't the hard part. The map is. *Click.*

Precision to hold. Eleven labs, 93,228 untreated wells, 1,872 plates, held out by plate; 78 wells wrong. Holding out whole batches gives 99.6%; labs on one

microscope model, 99.99%. The signature is in every feature family, image-quality features included, so it reflects the whole imaging pipeline as well as the cells. Say "the lab", not "the cells". Cell Painting fluorescence; brightfield is the bet.

For ML people. The line on the second view: normalising each plate to its own controls takes lab identity from 99.9 to 24 percent. Right for discovery, wrong for QC (A3).

03 Do my cells look normal? (0:46–1:11)

Felt. Point at the pink slice, then the six search results, then the two edges of card 3: run it, or hold.

0:46 So every lab asks the same question, and can't answer it. About twenty-eight billion dollars a year of US preclinical research can't be reproduced, and a third of that is inputs.

0:58 Today you search, you ask around, then you guess. I've seen one lab run too fast and one hold too long. Cells are the input nobody certifies. *Click.*

Precision to hold. \$28B is a 2015 US estimate; the 36% covers all reagents and reference materials, not only cells. The search card shows six real U2OS images from six papers, 2009 to 2023, via IDR and JUMP; sources are in the README. Card 3 names neither lab and gives no exact figure; keep it that way on stage. Growth chart is the metaphor if you need one: nobody plots their cells against the population.

04 Founder (1:11–1:36)

Lived. Let the logos and the pink line under your name do the credential work.

1:11 I've asked it at five benches. Nobody could answer it. *Click.*

1:17 At Axiom we onboarded hepatocytes through a good CRO. Their checks passed; ours passed. The first sign anything had moved came from outside: the

Broad couldn't align our profiles to their reference.

1:30 This is the thing I kept needing and nobody sold. *Click.*

Founder detail, for the judge question (about 30 seconds). Three methods, three papers, one thread. At AbbVie I built a neuronal tau aggregation assay for drug discovery [3]: the readout was what the cells did, not a label on them. At Axiom I built Cell Painting in primary human hepatocytes for cytotoxicity and mode of action [1], ran assay development for two of the largest Cell Painting datasets in the world, and invented two bioactivity detection methods that see a small molecule's effect on cells at about ten times lower concentration than LDH, ATP or cell count. With the Carpenter lab we showed that cell count alone predicts small-molecule bioactivity benchmarks [2]: even the simplest phenotype carries the signal. The thread is that bioactivity is a measurement of behaviour, and behaviour is the half of QC nobody measures. Every lab already images its control wells; Datum Line turns that image into the certificate, against the field.

Precision to hold. The question predates plates: at MGH it was whether a *Xenopus* oocyte, a single frog egg cell, looked normal before a recording. Say only what your author contribution statements support: "built" on [1] and [3], "we showed" on [2]. The two bioactivity methods and the ten-fold sensitivity are from your resume (macrofluidic.me/resume); if asked, name the comparison assays and the concentration range. The 2019 GABA_A paper [5] is the MGH anesthesia lab, Harvard Medical School, 2017 to 2018. The industry claim is the deck's own: STR and vendor certificates cover identity, bioactivity covers behaviour, and behaviour is what a control well measures in any lab that images one.

Precision to hold. "Published with" lists coauthor institutions, never partners. The Axiom story has no villain: the CRO passed every check it had, Axiom passed every check it had, and the failure was that no check for this existed. Name no vendor and no person. The uneven distribution was noticed; its severity and its effect on data quality were unknown, so never present it as the cause, and never say "spun down unevenly". If asked what it was: "We still don't know for certain. The uneven spreading

is one hypothesis; whole-well images show it more easily than our fields, and a field reference is what would tell you whether it matters." The sequence is: local checks passed, uneven spreading noticed but unquantified, the Broad flagged that reference profiles would not align, the cause is still open. Say what happened, not how it was fixed. Confirm with Axiom that you can tell it; if not, say "at one lab".

05 The map (1:36–1:54, two views)

The map. Dark slide: slow down. Two views, one sentence each.

1:36 Eleven labs, same cells, same protocol: eleven different normals. That's what the classifier saw. *Click.*

1:43 And where a lab sits on this map predicts how far its drug answers have moved. The map is worth having. *Click.*

Figure detail. First view: a PCA of 15,352 untreated wells for the eye; distances are measured in all features. The number is the mean distance between two labs' centres, 6.2 within-plate spreads. Second view: one point per batch; agreement is the cosine between a batch's effect of 7 positive controls and its lab's first-batch effect; rho is Spearman after removing each lab's mean, minus 0.71 over 118 batches; shuffled, minus 0.03; without lab 2, minus 0.68; drift and effect from different wells, minus 0.72. Lab 2 against its June answer: 0.85 before the flag, 0.67 after.

Precision to hold. In lab 2 the flagged batches are also the last three, so inside that lab the flag and the calendar cannot be separated; the correlation uses every lab. The ringed points are one lab's autumn batches (A13); the crosses are batches whose positive controls show no effect (A8). If asked whether the map changes a lab's results: yes, modestly. In the field's frame, agreement between labs on the same drugs goes from 0.75 to 0.78, above the usual normalisation to a batch's own controls (0.76) in 107 of 125 batches; it does not change a lab's agreement with itself (A14).

06 Every lab that joins sharpens the map (1:54–2:13)

Why it compounds. Point at the curve, then the three prices. This is the moat and the business model in one breath.

1:54 Put them on one map and it sharpens with every lab that joins: six apart with one lab, under three with five. The labs that build it use it free. The labs that cannot afford to be wrong pay, per line, per quarter. *Click.*

Precision to hold. The prices are a proposal, not a validated model: free for contributors, about 5 to 15 thousand dollars per line per quarter for screening groups, a fee per certified lot. One unchecked screen is about 2.6 million, so the subscription is about 170 times less (A15). Bad-lot alerts across labs come after the certificate; don't promise them.

Figure detail. Two references, each built from the same number of labs that share none; with one lab each they sit 6.2 within-plate spreads apart, with five each 2.8. The dashed line is one over root n . Eleven labs can make at most two disjoint sets of five, so the curve stops there. A lab's own baseline settles sooner: 0.89 with one plate, 0.39 with eight, then flat. The reference of all 11 labs moves 0.41 when one lab is removed.

07 Send your controls. See where you stand (2:13–2:36)

What. One sentence, then the map, then the certificate, then the row of ticks.

2:13 The product: send the control images you already take. They stay yours.

2:18 Back comes your place on the map, a certificate you can show, and data every other lab can compare. Every batch you send sharpens the map, for you and for everyone. It's a growth chart your cells come back to. *Click.*

Precision to hold. This is a replay on public data, not a customer. The same lab's later batches sit outside its baseline from 30 August, and the certificate says so and

says where to look: ER and AGP stain intensity (A11, A12, A13). Across the field 6 of 129 batches are outside. Bring that up if asked whether it ever says no. Brightfield is still the bet (A10).

Figure detail. One real lab on one real day: lab 2, 16 August, 25 plates, chosen as its in-distribution batch with the most plates. 2.8 spreads from its own baseline, 4.4 to lab 6, 188 cells per well. The reference is 1,652 plates from the other 10 labs. In the field's frame this batch's agreement with the other labs goes from 0.70 to 0.77. The ticks are the batches sent so far; the open one is the next.

08 Identity, yes. Behaviour, no. (2:36–2:46)

Point at the open tick in the Datum Line column: identity is done with a partner, STR. One sentence.

2:36 STR proves which line you have. Nothing proves it behaves like everyone else's. STR is a collaborator, not a rival. *Click.*

09 Eighteen months, four milestones (2:46–2:58)

The money. Four cards left to right; the pink one is the milestone that pays.

2:46 Eighteen months: three labs live on the map, brightfield proven, the ugliest dataset in biology built on purpose and graded, and a first line that pays. *Click.*

Figure detail. Milestones, not dates: the pre-seed size is not set and the proof of concept sets it. Three partner labs means three labs sending U2OS controls every batch and reading certificates; none has agreed yet, so say "three" and not a name. Brightfield proven means the same map from label-free images (A10). The test set is the ugliest dataset on purpose: six labs' real wells plus planted failures, bubble, focus, exposure, clipped well, bad lot, wrong label, with every method graded on naming the failure (A16). A first paying line is one pharma cell line on a per-quarter subscription.

10 Every plate, placed in the field (2:58–3:16)

Vision, then the ask. Point at the open row, then the right card. Stop on the question.

2:58 Every plate you image, placed on the map. Already there, from public data: 129 batches, eleven labs. The next row is yours.

3:07 What I need: images, U2OS first; ML people better than me; bench time; and a pre-seed.

3:13 Have images of U2OS cells? Talk to me.

Precision to hold. Price and test-set budget are not set: say "per line, per quarter" and "the proof of concept sets the size". Don't name a partner lab until one has agreed. Name no JUMP lab beyond its number. The grid is 129 batches and 1,871 plates; each square is a batch. Now is U2OS in fluorescence; more lines and brightfield are next and are not yet shown.